# Day 20 Assignment - 2

**Description:**
Built the foundational backend routes for the SkillSphere LMS using Express.js.
This day focused on basic routing, dynamic route parameters, and route-level validation middleware.

**Challenge 1 - Basic Route Setup:**
Created a root route to confirm that the SkillSphere LMS server is running correctly.
This established the base Express server setup.

server.js

```
const express = require("express");
const app = express();
const courseRoutes = require("./routes/courses");

app.use(express.json());
app.use("/courses", courseRoutes);

app.get("/", (req, res) => {
    res.send("Welcome to SkillSphere LMS API");
});

app.listen(4000, () => {
    console.log("Server running on http://localhost:4000");
});
```

localhost:4000

Welcome to SkillSphere LMS API

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Wipro\Assignments\Day 20\Day_20_Assignment_2_(skillsphere)> node .\server.js
Server running on http://localhost:4000
```

**Challenge 2 - Dynamic Routing:**
Built a dynamic course route using a URL parameter to return course details in JSON format.
This demonstrated how to access route parameters using req.params.

courses.js

```
const express = require("express");
const router = express.Router();

router.get("/:id", (req, res) => {
    const courseId = req.params.id;

    res.json({
        id: courseId,
        name: "React Mastery",
        duration: "6 weeks"
    });
});

module.exports = router;
```

localhost:4000/courses/101

Pretty-print

{"id":"101","name":"React Mastery","duration":"6 weeks"}

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Wipro\Assignments\Day 20\Day_20_Assignment_2_(skillsphere)> node .\server.js
Server running on http://localhost:4000
```

Saladi V S S Siva Hemanth Kumar
hemanthsaladi2004@gmail.com
WIPRO NGA - MERN - FY26 - Cohort 3

**Challenge 3 - Route Middleware for Validation:**

Added route-level middleware to validate that course IDs are numeric before processing the request. This improved input safety and maintainability.

courses.js

routes > courses.js > ...

```javascript
const express = require("express");
const router = express.Router();

function validateCourseId(req, res, next) {
    const courseId = req.params.id;

    if (!/^\d+$/.test(courseId)) {
        return res.status(400).json({ error: "Invalid course ID" });
    }

    next();
}

router.get("/:id", validateCourseId, (req, res) => {
    const courseId = req.params.id;

    res.json({
        id: courseId,
        name: "React Mastery",
        duration: "6 weeks"
    });
});

module.exports = router;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS D:\Wipro\Assignments\Day 20\Day_20_Assign
Server running on http://localhost:4000
```

localhost:4000/courses/101

localhost:4000/courses/101

Pretty-print

```
{"id":"101","name":"React Mastery","duration":"6 weeks"}
```

localhost:4000/courses/abc

localhost:4000/courses/hemanth

Pretty-print

```
{"error":"Invalid course ID"}
```

**Run Comands:**

npm init -y
npm install express
node server

Saladi V S S Siva Hemanth Kumar
hemanthsaladi2004@gmail.com
WIPRO NGA - MERN - FY26 - Cohort 3